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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

ATTENDANCE MONITORING SYSTEM

Submitted by

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DECLARATION

I hereby declare that the project entitled **"ATTENDANCE MONITORING SYSTEM"** submitted in partial fulfillment of the requirements for the course [B.Sc.IT I-A] is a record of my original work carried out under the guidance of [DHILEEBAN A], Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

CERTIFICATE

This is to certify that the project entitled **"ATTENDANCE MONITORING SYSTEM"** is a bona fide work carried out by **[AKSHAYA SRI A] [I A], [B.Sc.INFORMATION TECHNOLOGY]**, KG College of Arts & Science, in partial fulfillment of the requirements for the course [Course Name].

This project has been completed under my supervision and guidance during the academic period April 2026

Place:

Date:

Submitted for the Viva-Voce Examination held on _____

Guide:
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ABSTRACT

The Attendance Monitoring System is a software application built to streamline the process of managing and tracking student attendance in a fast, consistent, and reliable way. In many educational institutions, attendance is still recorded manually, which can quickly become tedious for instructors and administrative staff. Paper-based logs or spreadsheets are often prone to mistakes, missing entries, and delays in updating records. By offering an automated approach, this system reduces repetitive work, improves data accuracy, and makes attendance information available whenever it's needed.

The primary objective of the system is to record daily student attendance, calculate each student's overall attendance percentage, and determine eligibility based on predefined institutional criteria (such as a minimum required percentage). Every student is assigned a unique ID to ensure clear identification and to prevent duplicate or mismatched records. Attendance is updated on a regular basis so that percentages and eligibility decisions reflect the most current data.

To keep the solution organized and easy to use, the system is divided into three core modules: student registration, attendance marking, and report generation. The student registration module supports adding new students by collecting key details and storing them under a unique identifier. The attendance marking module allows the user to record whether a student is present or absent for a given session, ensuring that attendance status can be updated consistently over time. The report generation module produces clear summaries of attendance totals, percentages, and eligibility status, helping teachers and administrators quickly review performance and compliance.

1. INTRODUCTION

In today's modern educational settings, accurately tracking student attendance plays a vital role in maintaining academic discipline, supporting accountability, and helping schools evaluate overall student performance. However, many traditional attendance methods still depend on paper sheets or manual spreadsheets, which often create avoidable challenges. These approaches can be time-consuming for teachers, prone to human error, and vulnerable to misplaced records, inconsistent entries, and data loss. Over time, these issues can reduce reliability and make it harder to produce accurate summaries when they're needed most.

The Attendance Monitoring System is built to streamline and automate the entire attendance process, offering a simple, efficient, and dependable way to manage attendance records. With an easy-to-use interface, teachers and administrators can quickly record daily attendance, update entries when necessary, and generate clear reports without digging through manual files. Each student is assigned a unique identification number to ensure accurate tracking and prevent duplication. Attendance is updated on a daily basis, and the system automatically calculates attendance percentages in real time. It also determines whether each student meets required attendance thresholds, making compliance checks fast, consistent, and stress-free.

1.1 Objective

The main objective of the Attendance Monitoring System is to provide an efficient and automated method for managing student attendance.

- Reduce manual work
- Improve accuracy
- Calculate attendance percentage
- Determine eligibility (≥ 75%)
- Provide clear attendance reports

1.2 Overview

The system is a menu-driven application developed using Java. It allows users to add students, mark attendance, and view reports.

Each student record includes ID, name, total classes, and attended classes. The system calculates attendance percentage automatically and checks eligibility.

The application uses object-oriented programming and ArrayList for storing student data dynamically.

1.3 Features

Easily add new students to your system, then quickly mark attendance with a clean, menu-based interface that's simple to navigate. Generate and view detailed attendance reports anytime, with automatic percentage calculations built in.

The tool also performs an eligibility check based on a 75% threshold and includes reliable error handling, such as clear alerts when a student isn't found

2. PROGRAM ANALYSIS

The system evaluates common challenges found in manual attendance processes, including inaccurate or inconsistent records, excessive time spent on data entry, and the absence of reliable tracking and reporting.

To address these issues, it delivers an automated attendance solution that streamlines daily operations, improves accuracy, and minimizes human error. By digitizing and organizing attendance data in real time, the system helps organizations boost efficiency, maintain better oversight, and simplify record management.

2.1 Problem Definition

Manual attendance systems come with several common challenges that can slow teams down and create avoidable problems. They often require a time-consuming process for marking and compiling attendance, and they're prone to human errors in calculations or totals.
Maintaining accurate records over time can also be difficult, especially when documents are misplaced or inconsistent. On top of that, there's no automatic eligibility check, which leads to gaps and delays. These issues highlight the need for an automated attendance system.

2.2 Project Overview

The Attendance Monitoring System is designed to overcome manual limitations. It allows adding students, marking attendance, and generating reports.

The system calculates attendance percentage and determines eligibility automatically. It is implemented using Java and follows object-oriented programming principles.

2.3 Scopes and Limitations

Scope

- Track and manage student attendance efficiently for classes or small groups
- Automatically calculate attendance percentages based on entries
- Determine basic eligibility using clear, predefined rules
- Designed to be simple, quick to learn, and easy to use

Limitations

- Does not include database storage or long-term data retention
- No graphical user interface; runs via text-based interaction
- Attendance must be entered manually each time
- Not ideal for large institutions with high-volume or complex requirements

3. SYSTEM DESIGN

The system is divided into different modules to ensure better organization, easy maintenance, and smooth functionality. Each module performs a specific task and works together to achieve the overall system operation.

3.1 Modules

1. Menu Module

   The Menu Module displays all available options such as Add Student, Mark Attendance, View Report, and Exit. It helps users navigate through the system easily using a simple interface.

2. Student Management Module

   This module is responsible for adding new students and storing their details. It maintains student records efficiently using data structures like ArrayList.

3. Attendance Module

   The Attendance Module records whether a student is present or absent. It updates attendance data and keeps track of total and attended classes.

4. Report Module

   This module generates and displays attendance reports for all students. It shows attendance percentage and determines eligibility based on predefined criteria.

5. Exit Module

   The Exit Module terminates the program safely when the user chooses to exit. It ensures proper closure of the system without data loss.

4. METHODOLOGY

The methodology outlines the clear, step-by-step process used to plan, build, and validate the system. By following an organized approach, the project can be completed in a structured and systematic manner, with each phase producing outputs that guide the next stage. This reduces rework, improves consistency, and ensures the final solution meets the intended requirements for accuracy and usability.

- Problem Analysis
- System Design
- Program Development
- Testing and Validation
- Result and Documentation

4.1 Problem Analysis

In this phase, the key issues found in manual attendance tracking are examined in detail, including frequent human errors, slow processing time, misplaced or incomplete records, and overall inefficiency. The current workflow is reviewed to identify where mistakes commonly occur and why tracking becomes difficult as the number of students increases. Based on these findings, the need for an automated attendance system is analyzed, with a focus on improving accuracy, saving time, and enhancing performance. Core functional requirements, user expectations, and basic constraints are also identified to shape the solution.

4.2 System Design

The system design phase concentrates on planning how the application will be structured and how its main modules will interact. It defines the key components, data flow, and user-facing features needed for smooth operation. Object-oriented programming principles are applied to promote modularity, code reuse, and maintainability. This phase also considers how attendance data will be stored, accessed, updated, and displayed, ensuring a logical structure that supports future enhancements and easier debugging.

4.3 Program Development

During program development, the design is translated into a working system using the Java

programming language. Required classes and methods are created to represent students, attendance records, and related operations. Data structures such as `ArrayList` are used to store and manage student information dynamically, enabling efficient updates, searches, and record handling. This phase focuses on producing clean, functional code that supports reliable attendance management and aligns with the goals identified during analysis.

4.4. Testing and Validation

TEST CASE	DESCRIPTION	INPUT	EXPECTED OUTPUT	RESULT
TC01	Add Student	ID + Name	Student added	Pass
TC02	Mark Attendance	ID + true	Attendance updated	Pass
TC03	Invalid ID	Wrong ID	Student not found	Pass
TC04	View Report	-	Display details	Pass
TC05	Exit	4	Program exit	Pass

4.5. Result and Documentation

The Attendance Monitoring System was successfully developed and tested, and it performs all the required functions efficiently. The system allows users to add student details, mark attendance, and generate reports without any errors. It accurately calculates attendance percentage based on the number of classes attended and total classes conducted.

The results show that the system is reliable and user-friendly, as it provides correct output for all operations such as adding students, updating attendance, and displaying reports. It also determines student eligibility based on attendance criteria (75% or above), which helps in academic evaluation.

Proper documentation has been prepared for the system, including system design, algorithms, source code, and output results. This documentation helps in understanding the working of the system and makes it easier for future modifications and maintenance.

4.6. Algorithm

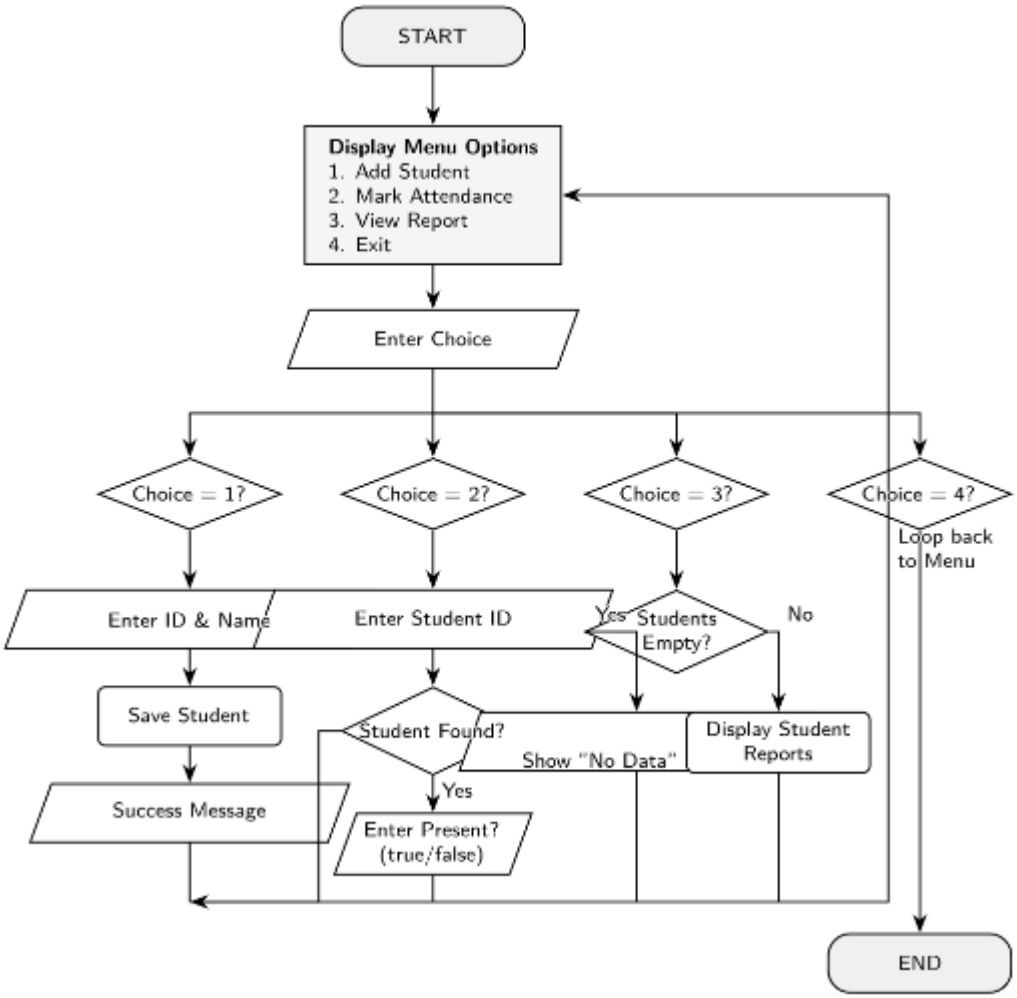
Algorithm: Add Student

1. Start
2. Enter ID and Name
3. Store in list
4. Display success message
5. Stop

Algorithm: Mark Attendance

1. Start
2. Enter student ID
3. Check if exists
4. If yes → mark attendance
5. Else → show error
6. Stop

4.7. Flowchart



4. SOURCE CODE

```
import java.util.*;

class Student {
    int id;
    String name;
    int totalClasses;
    int attendedClasses;

    Student(int id, String name) {
        this.id = id;
        this.name = name;
        this.totalClasses = 0;
        this.attendedClasses = 0;
    }

    void markAttendance(boolean present) {
        totalClasses++;
        if (present) {
            attendedClasses++;
        }
    }

    double getAttendancePercentage() {
        if (totalClasses == 0) return 0;
        return (attendedClasses * 100.0) / totalClasses;
```

```
    }

    boolean isEligible() {
        return getAttendancePercentage() >= 75;
    }

    void display() {
        System.out.println("\nID: " + id);
        System.out.println("Name: " + name);
        System.out.println("Attendance: " + attendedClasses + "/" + totalClasses);
        System.out.printf("Percentage: %.2f%%\n", getAttendancePercentage());
        System.out.println("Eligible: " + (isEligible() ? "Yes" : "No"));
    }
}

public class AttendanceSystem {
    static Scanner sc = new Scanner(System.in);
    static ArrayList<Student> students = new ArrayList<>();

    public static void main(String[] args) {
        int choice;

        do {
            System.out.println("\n--- Attendance Monitoring System ---");
            System.out.println("1. Add Student");
            System.out.println("2. Mark Attendance");
            System.out.println("3. View Report");
            System.out.println("4. Exit");
            System.out.print("Enter choice: ");
```

```
        choice = sc.nextInt();

        switch (choice) {
            case 1:
                addStudent();
                break;
            case 2:
                markAttendance();
                break;
            case 3:
                viewReport();
                break;
            case 4:
                System.out.println("Exiting...");
                break;
            default:
                System.out.println("Invalid choice!");
        }

    } while (choice != 4);
}

static void addStudent() {
    System.out.print("Enter ID: ");
    int id = sc.nextInt();
    sc.nextLine(); // consume newline

    System.out.print("Enter Name: ");
    String name = sc.nextLine();
```

```
    students.add(new Student(id, name));
    System.out.println("Student added successfully!");
}

static Student findStudent(int id) {
    for (Student s : students) {
        if (s.id == id) return s;
    }
    return null;
}

static void markAttendance() {
    System.out.print("Enter Student ID: ");
    int id = sc.nextInt();

    Student s = findStudent(id);
    if (s == null) {
        System.out.println("Student not found!");
        return;
    }

    System.out.print("Present? (true/false): ");
    boolean present = sc.nextBoolean();

    s.markAttendance(present);
    System.out.println("Attendance marked!");
}
```

```
    static void viewReport() {
        if (students.isEmpty()) {
            System.out.println("No students available!");
            return;
        }

        for (Student s : students) {
            s.display();
        }
    }
}
```

5. INPUT AND OUTPUT SCREENS

--- Attendance Monitoring System ---

1. Add Student

2. Mark Attendance

3. View Report

4. Exit

Enter choice: 1

Enter ID: 101

Enter Name: Rahul

Student added successfully!

--- Attendance Monitoring System ---

1. Add Student

2. Mark Attendance

3. View Report

4. Exit

Enter choice: 2

Enter Student ID: 101

Present? (true/false): true

Attendance marked!

--- Attendance Monitoring System ---

1. Add Student

2. Mark Attendance

3. View Report

4. Exit

Enter choice: 2

Enter Student ID: 101

Present? (true/false): false

Attendance marked!

--- Attendance Monitoring System ---

1. Add Student

2. Mark Attendance

3. View Report

4. Exit

Enter choice: 3

ID: 101

Name: Rahul

Attendance: 1/2

Percentage: 50.00%

Eligible: No

--- Attendance Monitoring System ---

1. Add Student

2. Mark Attendance

3. View Report

4. Exit

Enter choice: 4

Exiting.

6. DISCUSSION & CONCLUSION

6.1 Discussion

The Attendance Monitoring System developed in this project provides a simple and efficient solution for managing student attendance. It eliminates the need for manual record-keeping, thereby reducing human errors and saving time. The system ensures accurate calculation of attendance percentage and helps in identifying student eligibility based on predefined criteria.

The modular design of the system makes it easy to understand, use, and maintain. Each module performs a specific function such as adding students, marking attendance, and generating reports, which improves the overall efficiency of the system. The use of object-oriented programming concepts enhances code reusability and scalability.

However, the system has certain limitations. It currently operates as a console-based application and requires manual input for attendance marking. It does not store data permanently, as it lacks database integration. Additionally, it is more suitable for small-scale use and may require enhancements for handling large numbers of students.

6.2 Conclusion

The Attendance Monitoring System is a useful and effective application that simplifies the process of managing student attendance. It automates important tasks such as recording attendance, calculating percentages, and determining eligibility, which improves accuracy and reduces manual effort.

The system is designed using Java and follows object-oriented principles, making it flexible, reliable, and easy to maintain. It successfully meets the basic requirements of an attendance management system and provides a strong foundation for further development.

In the future, the system can be enhanced by integrating a database for permanent data storage, adding a graphical user interface (GUI) for better user experience, and including advanced features such as automated attendance tracking, report generation, and online access. These improvements can make the system more powerful and suitable for real-world applications.

7. REFERENCES

The following resources were referred to during the development of the Attendance Monitoring System project. These materials helped in understanding Java programming concepts, object-oriented principles, and system design techniques.

Books

Herbert Schildt – Java: The Complete Reference

This book was used as a primary reference for understanding core Java concepts, including classes, objects, methods, and exception handling. It also provided detailed explanations of object-oriented programming.

E. Balagurusamy – Programming with Java

This book helped in learning fundamental and advanced Java programming techniques. It provided clear examples and explanations that were useful for implementing the project.

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Websites

Oracle Documentation

🔗 Visit Oracle Java Official Site

→ Official source for Java programming, tools, and documentation.

Oracle

GeeksforGeeks

🔗 Visit GeeksforGeeks Java Tutorial

→ Provides detailed tutorials on Java, data structures, and programming concepts.

GeeksforGeeks

W3Schools

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→ Beginner-friendly platform with simple examples and exercises for learning Java.

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